Unit-4 - Biomass

Biomass, Sources of biomass, Pyrolysis, combustion and gasification process, Updraft and downdraft gasifier, Fluidized bed gasifier, Fermentation and digestion process, Fixed and floating digester biogas plants, Operation of biogas plants, Applications

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What is Biomass?

- All living and recently living organisms, animal/plant waste, industrial and municipal waste
- Biomass is organic material from plants and animals that can be used as a renewable energy source

- Categories
  - Bio fuels
  - Bio power
  - Bio products
  - Bio refineries

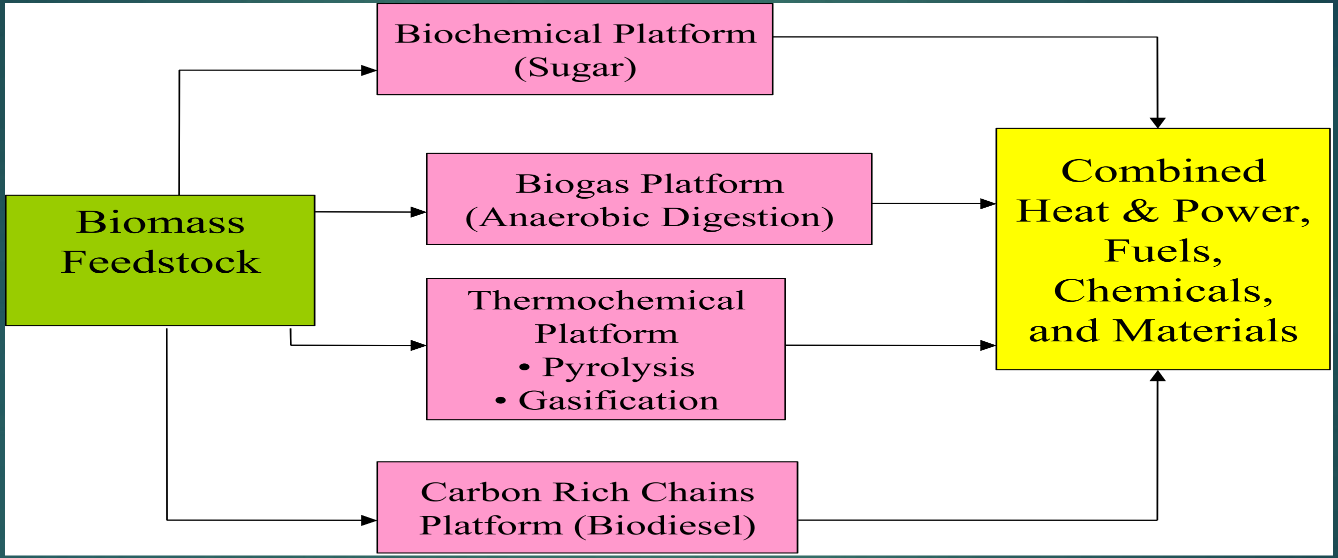
Sources of biomass

Sources of biomass include :

- agricultural residues like crop waste,
- forestry residues such as wood and bark,
- dedicated energy crops grown specifically for energy,
- animal waste and sewage,
- algae and aquatic plants,
- the organic portion of municipal solid waste.

These organic materials store solar energy and can be used to generate electricity, heat, or biofuels.

Conversion Technologies



Biochemical Conversion

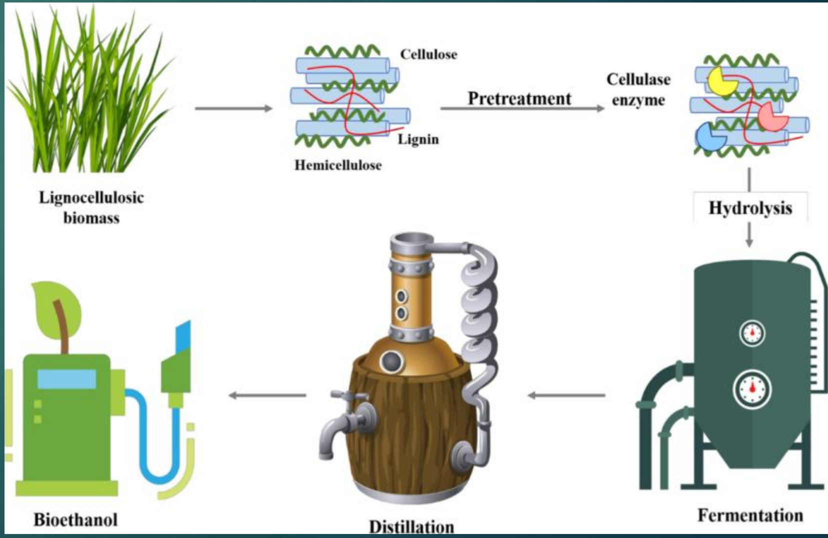
Biochemical conversion uses enzymes and microorganisms to transform organic materials like biomass into biofuels, chemicals, and other valuable products.

- Plant matter – hemicellulose, cellulose, lignin
- Pretreatment
- Hydrolysis
- Sugar Fermentation

Cellulose forms strong, linear fibers, while hemicellulose is a shorter, branched polymer that helps create a matrix. Lignin is a complex, aromatic polymer that adds rigidity and strength, acting as a binding agent for the other components.

$C_{12}H_{22}O_{11}$	+	H_2O	Invertase →	$C_6H_{12}O_6$	+	$C_6H_{12}O_6$
Sucrose		Water	Catalyst	Fructose		Glucose

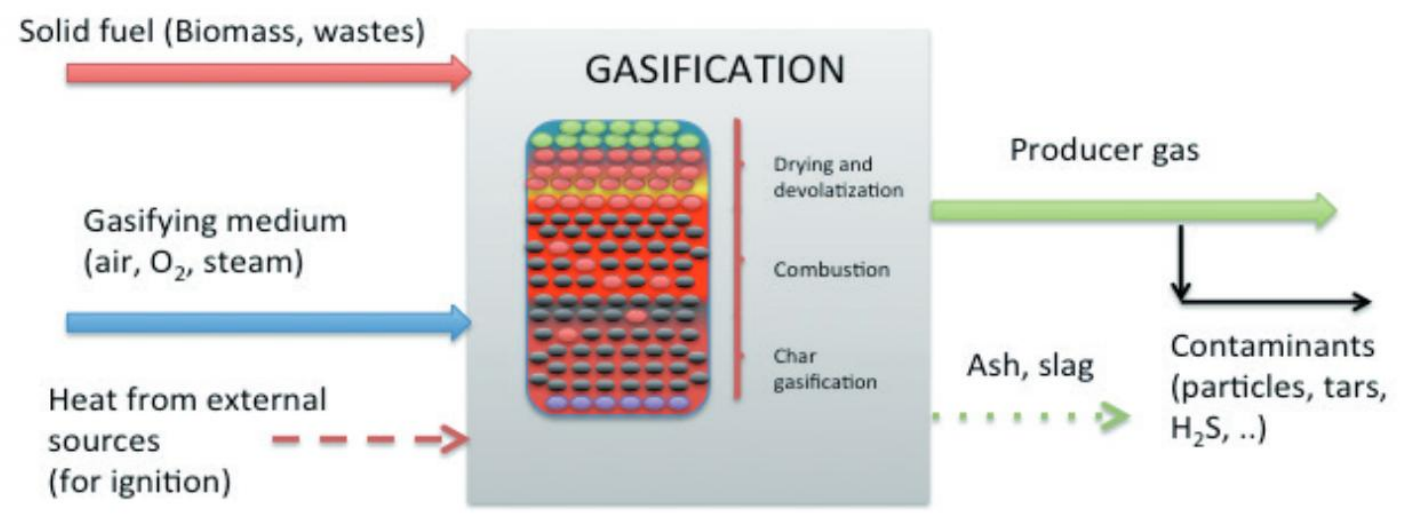
$C_6H_{12}O_6$	Zymase →	$2C_2H_5OH$	+	$2CO_2$
Fructose / Glucose	Catalyst	Ethanol		



Thermochemical Conversion

- Thermochemical conversion uses heat to break down biomass (organic matter) into fuels and chemicals by employing processes like combustion, gasification, and pyrolysis.
- Combustion burns biomass for heat, gasification converts it to syngas, and pyrolysis transforms it into solid char, liquid bio-oil, and gases by heating it without oxygen.

gasification



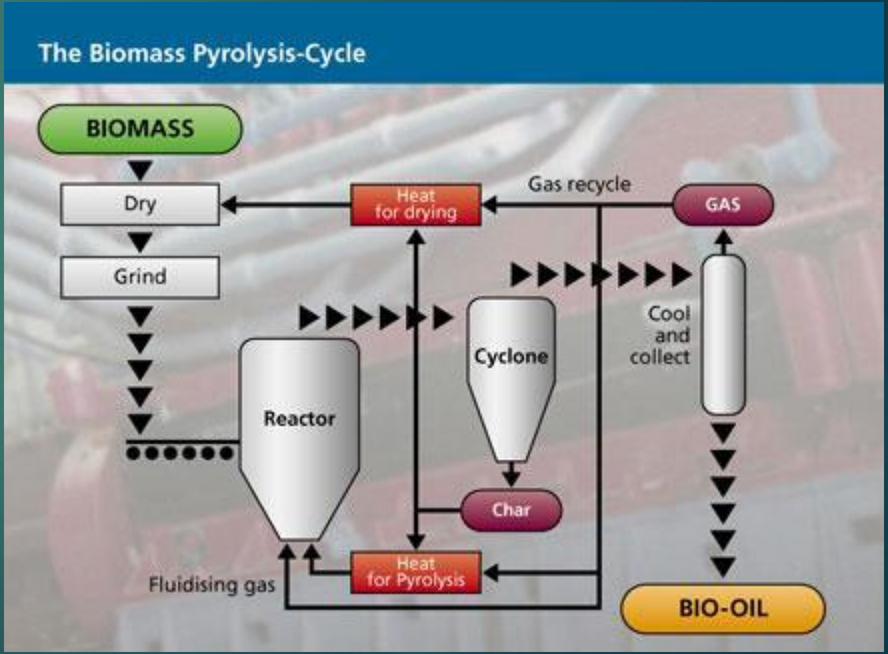
Pyrolysis

Pyrolysis is the thermochemical decomposition of organic materials at high temperatures in an oxygen-free environment, breaking down complex molecules into simpler ones like gases, bio-oil, and solid biochar

- Absence of oxygen →Thermal degradation →Liquid pyrolysis oil

How Pyrolysis Works:

- Heating: Organic material, such as waste plastics or biomass, is heated to temperatures typically ranging from 300–900 °C.
- Oxygen-Free Environment: The process occurs in the absence of oxygen, or with only a very limited amount, to prevent complete combustion.
- Decomposition: The heat breaks the large, complex organic molecules into smaller, simpler ones.
- Product Formation: The resulting products are a mix of gases, a liquid called bio-oil, and a solid residue known as biochar.

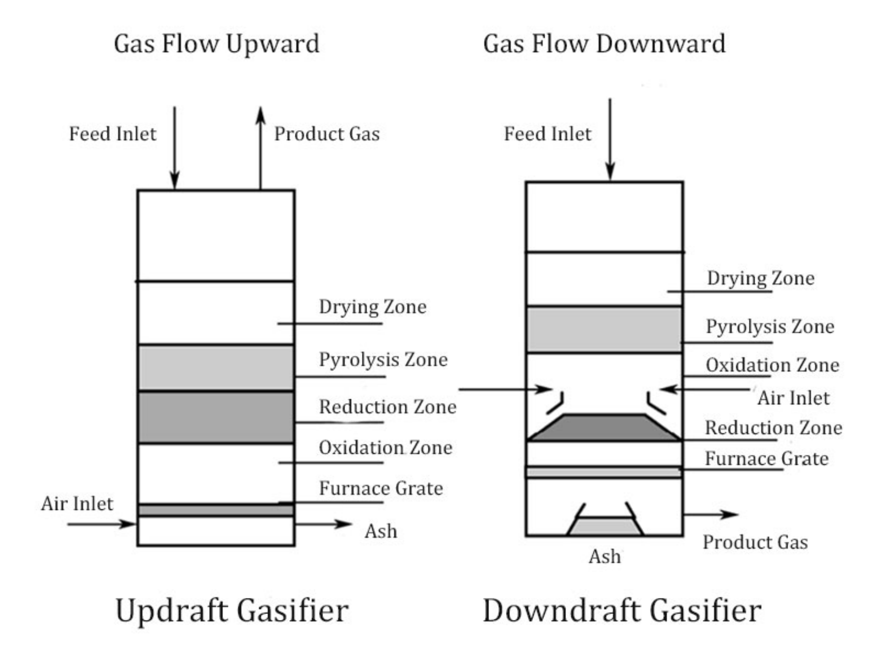


Updraft and downdraft gasifier

Updraft and downdraft gasifiers differ in their fuel and air flow directions and the quality of the syngas produced.

Updraft gasifiers use a counter-current flow, where air enters from the bottom and fuel from the top, producing a high-tar, low-temperature gas ideal for heating applications but unsuitable for engines.

Downdraft gasifiers, with their co-current flow where both fuel and air move downward, crack tar in the hot zone, yielding a cleaner, high-temperature gas suitable for thermal power generation and engines, though they are less fuel-flexible and have a lower thermal efficiency



Updraft Gasifier (Counter-Current)

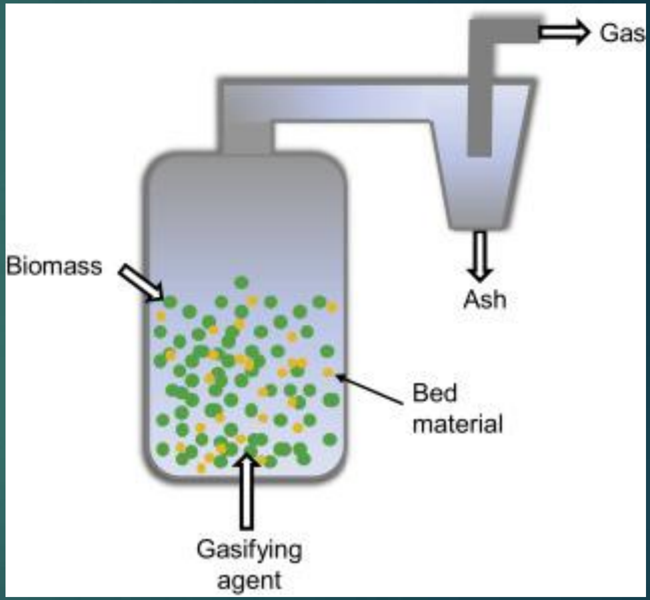
- Flow: Counter-current flow, with air and other gasifying agents injected from the bottom and fuel moving from the top under gravity.
- Gas Quality: Produces a low-quality gas with high temperatures (at the bottom) but low-temperature exiting gases.
- Tar Content: High in tar, making the gas unsuitable for engines.
- Efficiency: Thermally efficient due to the ascending gases drying and pyrolyzing the incoming biomass.
- Applications: Suitable for heating applications and uses fuels with high moisture and ash content, like biomass and coal.
- Limitations: The high tar content requires significant downstream cleaning for many applications.

Downdraft Gasifier (Co-Current)

- Flow: Co-current flow, where fuel and gasifying agent (air) move in the same direction, downward.
- Gas Quality: Produces a cleaner gas with low tar content because the tars are cracked in the hot hearth zone.
- Tar Content: Low tar content, but higher particulate content.
- Efficiency: Lower overall thermal efficiency because the gases exit at high temperatures (800–1000°C), removing a significant amount of heat.
- Applications: Produces a clean, high-heating-value gas suitable for engines and power generation.
- Limitations: Less fuel-flexible and requires well-defined fuels (e.g., pelletized or briquetted biomass).

Fluidized bed gasifier

- A fluidized bed gasifier is a gasification system that suspends fuel particles in a semi-suspended, fluid-like state using an upward flow of a gasifying agent like air or steam through a bed of inert granular solids, such as sand.
- This process creates excellent gas-solid mixing, heat transfer, and temperature uniformity, leading to efficient gasification of various fuels, including low-grade coals and biomass.
- The key advantages are its ability to handle fuels with high ash content, lower operating costs due to no pre-processing, and improved fuel conversion rates.



Fluidized bed gasifier

How it Works

- **Fluidization:** A gasifying medium (air, steam, or oxygen) is forced upwards through a bed of granular material (like sand) at high pressure.
- **Suspension:** The fuel particles are introduced and become suspended within this bed, behaving like a fluid.
- **Mixing:** This fluid-like state ensures excellent mixing between the fuel, bed material, and gasifying agent, promoting efficient heat and mass transfer.
- **Gasification:** At temperatures typically between 700–1000°C, the fuel undergoes partial oxidation, breaking down into a mixture of gases, primarily carbon monoxide (CO) and hydrogen (H_2).

BIODIGESTER

- Biogas Platform - Methane
- Decomposition - microorganisms
- Anaerobic Digesters
- Uses wastes and turns into valuable compost

- The biodigester is a physical structure, commonly known as the biogas plant.
- Since various chemical and microbiological reactions take place in the bio-digester, it is also known as bio-reactor or anaerobic reactor.
- The main function of this structure is to provide anaerobic condition within it.
- As a chamber, it should be air and water tight.
- In this three main process occur. (Hydrolysis, Acid formation , Methane formation)

Hydrolysis

- The complex organic molecules like fats, starches and proteins which are water insoluble contained in cellulosic biomass are broken down into simple compounds with the help of enzymes secreted by bacteria.
- This stage is also known as polymer breakdown stage (polymer to monomer).
- The major end product is glucose which is a simple product.

Acid formation

- The resultant product (monomers) obtained in hydrolysis stage serve as input for acid formation stage bacteria.
- Products produced in previous stage are fermented under anaerobic conditions to form different acids.
- The major products produced at the end of this stage are acetic acid, propionic acid, butyric acid and ethanol.

Methane formation:

- The acetic acid produced in the previous stages is converted into methane and carbon dioxide by a group of microorganism called “Methanogens”.
- In other words, it is process of production of methane by methanogens.
- Methanogens utilise the intermediate products of the preceding stages and convert them into methane, carbon dioxide, and water.
- It is these components that make up the majority of the biogas emitted from the system.
- Methanogenesis is sensitive to both high and low pH’s and occurs between pH 6.5 and pH 8.

Biogas Plant and Its Components

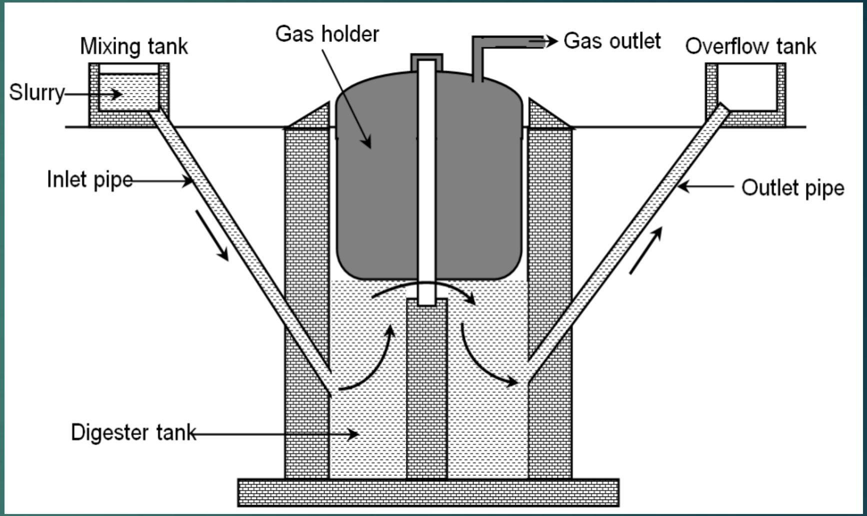
- • Mixing Tank: Cow dung is mixed with water (1:1) to make slurry.
- • Feed Inlet Pipe/Tank: Slurry enters digester through inlet.
- • Digester: Anaerobic fermentation by microorganisms.
- • Gas Holder: Stores biogas (drum in KVIC, dome in Janata plants).
- • Slurry Outlet Tank/Pipe: Digested slurry exits digester.
- • Gas Outlet Pipe: Releases and conveys stored gas.

Classification of biogas plant:

- • Floating drum type – KVIC model
- • Fixed dome type model – Deenbandhu model

FLOATING DRUM TYPE (CONSTANT PRESSURE)

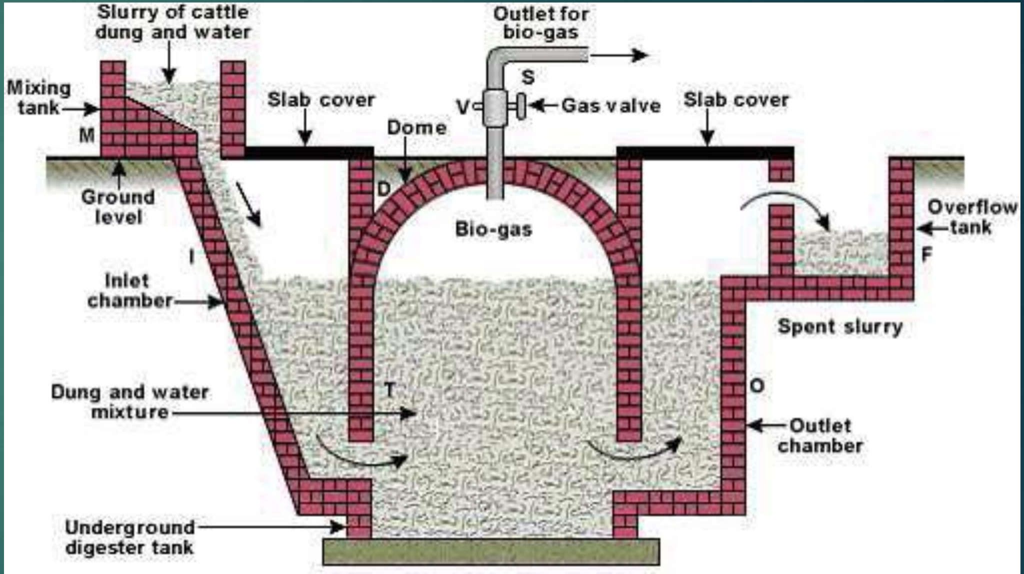
- • The floating gas holder type bio gas plant consists of a dome shaped gas holder made of steel for collecting bio gas.
- • The dome shaped gas holder is not fixed but is moveable and floats over the slurry present in the digester tank.
- • Due to this reason, this biogas plant is called floating gas holder type biogas plant.



FLOATING DRUM TYPE

Fixed Dome Type (CONSTANT VOLUME)

- • This type of biogas plant is very economical is design. It works with the constant volume principle.
- • The main structure is made up of brick and cement masonry. This type of plant doesn't have any moving parts so it is safe from wear and tear.
- • The operating pressure varies from 0 to 100 cm of water column.
- • It is also known as Janata model.



Fixed Dome Type:

Advantages

- • It has low cost compare to floating drum type, as it uses only cement and no steel.
- • It has non-corrosion trouble.
- • It this type heat insulation is better as construction is beneath the ground.
- • Temperature will be constant.
- • Cattle and human excreta and long fibrous stalks can be fed. • No maintenance.

Disadvantages:

- • This type of plant needs the services of skilled masons, who are rather scarce in rural areas.
- • Gas production per cum of the digester volume is also less.
- • Scum formation is a problem as no stirring arrangement
- • It has variable gas pressure

Operation of Biogas plant :

- • Slurry is prepared by mixing water in cattle dung in equal proportion in mixing tank.
- • The slurry is then injected into a digester tank with the help of inlet pipe.
- • The digester tank is a closed underground tank made up of bricks. Inside the digester tank, the complex carbon compounds present in the cattle dung breaks into simpler substances by the action of anaerobic microorganisms in the presence of water.
- • This anaerobic decomposition of complex carbon compounds present in cattle dung produces bio gas and gets completed in about 60 days.
- • The bio gas so produced starts to collect in floating gas holder and is supplied to homes through pipes. And the spent slurry is replaced from time to time with fresh slurry to continue the production of bio gas.

Applications of a biogas plant:

- **Cooking Fuel** – Biogas can replace LPG, kerosene, and firewood for household and community cooking.
- **Electricity Generation** – Biogas can be used in biogas engines or generators to produce electricity.
- **Heating Applications** – Used for space heating, water heating, and industrial thermal applications.
- **Lighting** – Biogas lamps can provide illumination in rural and remote areas.
- **Transport Fuel** – After purification (removing CO_2 and H_2S), biogas can be upgraded to **bio-CNG** and used in vehicles.
- **Agricultural Use** – The digested slurry (bio-slurry) is a nutrient-rich organic fertilizer that enhances soil fertility.
- **Waste Management** – Helps in safe disposal of animal waste, kitchen waste, and other biodegradable waste.
- **Environmental Protection** – Reduces deforestation (less firewood use), lowers greenhouse gas emissions, and controls odor and pathogens.

Thank

You